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ScienceDirect

Procedia Computer Science 233 (2024) 464–473

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

A Deep Learning-based Smart System for Ornaments and Devices Detection during Check-In of Entrance Exams

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Abstract

Nowadays, individuals are experiencing an era characterized by a pandemic situation. Given the challenging circumstances face these days, it is imperative to develop a real-time model that automatically detects whether candidates are wearing ornaments or carrying any devices before appearing for an entrance examination, as it serves as a vital precautionary measure for our health and safety. The proposed system will function in two stages: firstly, it will employ the YOLOv7 algorithm, which operates on Convolutional Neural Networks (CNN), to detect any gadgets or ornaments worn by the candidate. Additionally, we incorporated Mobile Net and HairNet into the code, utilizing TensorFlow's Keras API. Notably, all the datasets used in this system have been created from scratch. Upon successfully detecting gadgets such as wristwatches, Bluetooth devices, earrings, and other materials, the system will promptly display a warning message, denying entry to the respective individuals. Moreover, the proposed system includes real-time monitoring of the examination hall through CCTV cameras. As a cost-effective, time-saving, and accurate solution, it can be implemented across various examinations to ensure candidates' and invigilators' safety and adhere to examination norms.

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Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Convolutional Neural Networks (CNN); Real-time object detection; Deep learning

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10.1016/j.procs.2024.03.236

1. Introduction

In today's technology era, the increasing automation, particularly through the utilization of CCTV cameras, enables the identification of individuals at the entrance to various public places, including malls, airports, railway stations, marts, and more, without the need of manual intervention. However, when it comes to examinations, each student must still be individually checked and adhere to the specific rules and protocols set for exams such as the Gate, JRE, NEET, JEE-Mains, and admission exams of prestigious universities. Despite the advanced capabilities of the proposed model, it may face challenges in detecting non-metallic ornaments and devices, accurately making manual inspection necessary for identifying such items. To address this challenge, we have undertaken the development of cutting-edge software that aims to ensure strict adherence to examination rules by students without requiring any manual intervention. This paper deals with a novel model designed to detect prohibited items such as wristwatches, Bluetooth devices, earrings, and individuals with open hair, leveraging camera technology. The model will promptly display a warning message of 'ENTRY DENIED' for any candidate found possessing such items during the examination process, thereby bolstering the integrity and security of the exams. This paper delves into the evolution of an innovative model, showcasing its potential to revolutionize examination security. Examining real-world scenarios and illustrative examples, we unveil how this novel technology ensures stringent adherence to examination rules, eliminating the necessity for manual intervention and bolstering the integrity of academic assessments.

2. Research Focus

2.1. Ongoing Initiative

Until now, examination centers like JEE Mains, Gate, etc., have relied solely on manual checking or manual scanning, lacking an automated detection system. Furthermore, even in online exams, there is a lack of a comprehensive detection system that ensures all students are adhering to examination norms. With the onset of the COVID-19 pandemic, manual detection of gadgets and prohibited items pose a risky situation, emphasizing the need for an automated solution for enhanced safety and security during examinations. Enhancing the examination security detection system via video capture requires a systematic approach to problem formulation. This entails defining metrics to gauge the frequency of unauthorized items during exams, utilizing statistical data from past incidents and academic studies.

2.2. Project Pitch

The forthcoming experimental work aims to enhance the methodology by incorporating a more comprehensive quantitative framework, encompassing dataset metrics, training parameters, performance metrics, and real-world scenario simulations. These meticulously chosen metrics will not only ensure transparency and reproducibility but also contribute to a more rigorous evaluation of the proposed system. The study will additionally entail a quantitative comparison with existing models, evaluating speed, accuracy, and false positive rates for a thorough assessment. The envisioned system will operate in two distinct stages. Initially, it will employ advanced algorithms to detect any presence of gadgets, ornaments, or open hair by the candidates. The system's algorithms will be capable of identifying items such as analogue watches, smartwatches, Bluetooth devices, earrings, and bracelets. Once the detection process is complete, the integrated system will promptly issue a warning message of 'ENTRY DENIED' if any prohibited items are found. Conversely, candidates without any detected violations will be permitted to enter the exam hall without hindrance.

3. Pamphlet

3.1. Real-time Facemask Identification using Deep Learning

The real-time facemask identification system approach mainly contains two steps, the first one is the offline step and the second one is the online one. In the first step a deep learning design is created, and then it can detect and locate facemasks also it will detect if it is worn properly. The next one online step, which involves the deep learning model for edge computing infrastructure, enabling real-time mask detection. In their comparison, they evaluated the MobileNetV2 CNN model alongside other deep learning models, including DenseNet [24], ResNet50 [27], and VGG16. The results demonstrated that MobileNetV2 outperformed the others regarding training efficiency, achieving lower loss and higher accuracy, specifically reaching a remarkable 99% accuracy. Consequently, the author recommended the utilization of MobileNetV2 for real-time facemask detection. However, during the initial testing of the trained model on the first version of the dataset, a significant number of inaccurate results were observed due to conflicting images.[1]. As a result they persisted in this iterative process until they finally reached a satisfactory outcome, which was eventually accomplished with the eighth iteration of the dataset. To minimize the occurrence of both false positives and false negatives during execution, they introduced a test to confirm that four consecutive frames in the video consistently depicted whether the person was wearing a mask or not. This ultimately led to a significant enhancement in the accuracy of discerning whether a person was wearing a mask, regardless of the viewing angle.[2].

3.2. Real-time Facemask Identification using Deep Learning This study employs Deep Learning techniques to introduce a method for segmenting hair and extracting color information from facial images.

This paper delves into the process of segmentation of image pixels into distinct classes, including hair, faces, and backgrounds, employing a neural network that is fully convolutional and designed for multitasking. The proposed system initially employs a dlib library face detector to identify faces within the image, and additionally, it evaluates whether an individual has a bald head or not. The model defines a region of interest (ROI) around each detected face using the face detector in the dlib library. Following this, the cropped image is then passed to the segmentation module for further processing. In the segmentation module, each pixel is assigned to one of the pre-defined classes: hair, facial skin, or background. This is achieved through the combined utilization of U-Net and VGG architectures. Additionally, the module also assesses whether the person shown in the image is bald or not. If the individual is not bald, the image along with its corresponding segmentation mask is then forwarded to the hair color recognition module. This module employs the SLIC (Simple Linear Iterative Clustering) algorithm to identify the hair color, categorizing it into one of the following options: brown, grey/white, black, blonde, brown, or red. [3][4].

3.3. This research focuses on achieving real-time hair segmentation using a mobile U-Net model

The article introduces an innovative approach to real-time hair segmentation, leveraging the Mobile-U-Net, a deep-learning model specifically optimized for mobile devices. This method not only attains exceptional accuracy but also delivers real-time performance, rendering it well-suited for diverse applications, including virtual try-on and augmented reality. In their method evaluation, the authors employed the PASCAL VOC 2012 dataset, a widely recognized benchmark for image segmentation. Their approach was compared to other state-of-the-art models regarding both accuracy and execution speed.

The suggested approach holds promise for diverse sectors, including virtual reality, augmented reality, the gaming industry, and the beauty sector. To illustrate, it can enable users to experiment with various hairstyles or hair colors in a virtual try-on scenario without altering their actual hair. In augmented reality, virtual hairstyles can seamlessly blend with the real world, offering interactive and immersive experiences. Within the beauty industry, the suggested approach finds its utility in virtual hair color and hairstyle try-ons. In essence, the article offers valuable perspectives on the advancement of real-time hair segmentation technology through the utilization of deep learning models. The proposed method holds the promise of improving various applications demanding precise and swift hair segmentation on mobile devices. [5]

3.4. Real-time Facemask Identification using Deep Learning Utilizing a Convolutional Neural Network (CNN) for the detection of face masks as a measure to mitigate the transmission of COVID-19.

The study showcases the efficacy of three deep learning approaches in the realm of face mask recognition, specifically utilizing the MobileNetV2 architecture, along with Max pooling and Average pooling techniques. In-order to train the deep learning model, a dataset consisting of 1845 images from various sources, as well as 120 photographs of the co-authors captured using both a webcam and a mobile camera, was employed.[6] The performance results were as follows: Max pooling achieved a validation accuracy of 98.67% and a training accuracy of 96.49%, while average pooling yielded a validation accuracy of 96.23% and a training accuracy of 95.19%. Moreover, the MobileNetV2 architecture exhibited outstanding performance, achieving a training accuracy of 99.72% and a validation accuracy of 99.82%. [7]

3.5. Real-time Facemask Identification using Deep Learning Utilizing Deep Learning for the Detection of Face Masks

This process involves three key stages: preprocessing, CNN training, and finalization. In real-time classification, the proposed CNN is responsible for distinguishing between faces wearing masks and those without them. This classification is achieved through the utilization of two output neurons with SoftMax activation in the CNN's output layer. The loss function employed is categorical cross-entropy, and the suggested model achieves a validation accuracy of 96%. In the video stream, individuals not wearing protective masks have a red rectangle labeled "NO MASK" drawn around their faces, while those wearing masks have a green rectangle labeled "MASK" drawn around their faces. [8]

4. Inception Enactment

4.1 Data build-up

The technique of adding metadata, labels, or tags to a dataset to provide the dataset with more details or to allow machine learning algorithms to learn from the dataset is termed dataset annotation. The general procedures for dataset annotation are as follows: Annotation: Defining the annotation task and determining the kinds of annotations needed are the first steps. This involves deciding on the categories for the labels, the sort of material that needs to be annotated, and any other data that must be recorded. Assemble the data set: The next step is to compile the dataset after the annotation task has been defined. This entails gathering information from numerous sources and putting it in an easily annotatable style.

Choose an annotation tool: There are many annotation tools that can be used for dataset annotation, including Labelbox, Doccano, and Prodigy. Select a tool according to the requirements. Execute the annotating procedure: According to the annotation rules, the annotation process entails adding the necessary metadata, labels, or tags to the dataset. To guarantee accuracy and consistency, this procedure should be overseen, and quality control mechanisms should be put in place.

Verify the annotated text: It’s crucial to evaluate the annotations after they’ve been finished to make sure they’re exact and consistent. To do this, a sample of the annotations will be examined and compared to the annotation standards. The dataset should be cleaned and formatted once the annotations have been verified to make sure that it has been set up for application in machine learning techniques Use the annotated dataset: You may now use the annotated dataset for a few things, including training machine learning models.

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4.2 Architecture

4.2.1 YOLO V7 Model

In a YOLO [28] model, a backbone processes image frames, and these features are subsequently combined and fused within the network's neck. From there, they are directed to the model's head, where YOLO makes predictions regarding the positions and categories of objects for which bounding boxes should be created. [9][10]. Fig. 1 depicts the working model framework. To optimize data in YOLO object detection, one can employ various strategies, such as subset selection, where a representative data set is randomly sampled to ensure a diverse array of object instances. Additionally, techniques like image augmentation, object filtering, class balancing, data cleaning, transfer learning, and pruning can be implemented to enhance the efficiency of the model. Image augmentation boosts dataset variability by introducing diverse transformations without the need for additional data collection. Object filtering selectively removes less relevant instances, while class balancing ensures an even distribution of classes, contributing to a more robust and effective model in YOLO object detection. Data cleaning is employed to eliminate noisy or irrelevant images, streamlining the dataset for enhanced model performance. Transfer learning then comes into play, fine-tuning a pre-trained model specifically on this refined dataset, leveraging the knowledge gained from the broader data reduction process in YOLO object detection.

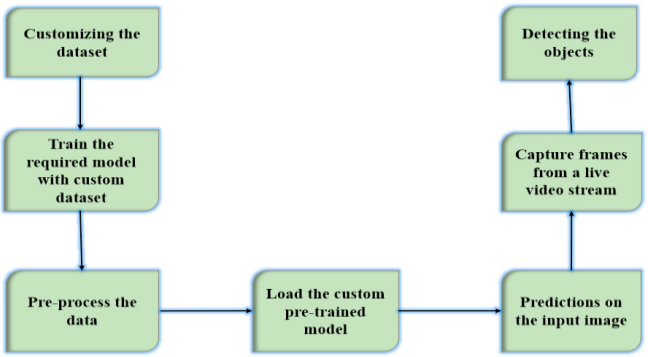


Fig. 1. Working model framework

4.2.2 Expanding the Efficiency of Layer Aggregation

The efficiency of the convolutional layers within YOLOv7's backbone plays a crucial role in achieving fast inference speeds. This efficiency considers both the memory needed to store layers and the distance over which gradients need to back-propagate across these layers. [11] [12]. A shorter gradient distance enhances the learning capacity of their network. E-ELAN, an extended version of the ELAN computational block, serves as the final layer aggregator. [9]

4.2.3 Techniques for Scaling Models

Object detection models are typically provided in a range of sizes to accommodate the diverse requirements of different applications, which may necessitate varying levels of accuracy and inference speed. These models consider factors such as network depth, network width, and the resolution used for training.[9]

4.2.4 Planning for Re-parameterization

In re-parameterization strategies, the process of averaging a set of model weights is employed to create a model that exhibits greater robustness to the specific patterns it's designed to capture. Recent research has placed significant focus on module-level reparameterization, wherein each component of the network possesses its own re-

parameterization algorithms.[9]

4.2.5 Coarse-to-Fine Auxiliary Head

While the YOLO network head delivers the ultimate network predictions, considering its location deep within the network, it can be advantageous to incorporate an auxiliary head at various intermediate points within the network. Throughout the training process, the auxiliary head oversees both the detection and prediction heads. However, due to the reduced network distance between the auxiliary head and the prediction head, it may not undergo training with the same level of efficiency as the final head.[9].

5. Work Breakdown

YOLOv7 does not rely on pre-trained weights like ImageNet's backbone, in contrast to earlier YOLO versions that did. [13] [28] As a result, the model learns features that are particularly customized to the data at hand rather than relying on pre-existing weights from a separate dataset, making it more adaptive to smaller datasets.[14] [15].

5.1 Customizing the Dataset

Firstly, to train YOLOv7 on a custom dataset, we must first clone the YOLOv7 repository and install the required packages, swap out torch and torch vision for CUDA-enabled versions. Then, we need to move our images and annotations to their appropriate folders for training and validation purposes. The images must be kept in a folder called images and annotations in the labels folder. Later, we must move the remaining 20% to the val folder and the remaining 80% of the photographs and labels to the train folder.

5.2 Training the required Model with Custom Dataset

Open coco.yaml, then set train: data/train and val: data/val. We also need to set the number of classes and names, which should be 6 as our project has six classes with the names ['ear buds','earrings','analogwatch','bracelet','smartwatch','open hair']. In YOLO setup, the number of classes must also be specified. After all of this, we must download Yolov7 per-trained weights and test the model.

Yolov7 must be trained using the custom dataset, hence we must set the runtime to GPU. Install Google Drive on Google Collaborate. This will direct you to a page where you may request an authorization. Please click the link to provide Collab access to your Google Drive account. In Co-lab, your Google Drive will be mounted. Use the %cd command to get to the specified directory. We must confirm that PyTorch is set up and operating on a computer with a CUDA-capable GPU.

5.3 Pre-process the Data and Load the Custom Pre-trained Model

Define letterbox, it accepts an image as input and, while adhering to stride-multiple limitations, re-sizes and pads the picture. The goal of the function is to guarantee that an image may be processed by a model that needs a specific input size and stride. The function makes sure that the picture has the proper size and that the stride criteria are satisfied by scaling and padding the image. The scaling ratio between an image's new and old forms must then be determined. The minimal ratio between the new height/width and the previous height/width is used to compute the scaling ratio.[16] As a result, the full image may be accommodated inside the new proportions. This method seems to calculate the amount of padding necessary to resize a picture while preserving the original image's aspect ratio. The function accepts a picture as a NumPy array, the desired output shape, the scaling ratio, a Boolean value indicating whether to stretch the image to fill the output shape or automatically calculate the smallest rectangle to fit the image within and a color value for the padding border. To apply object detection to an input picture supplied by the source image path variable, the YOLOv7 model must first be loaded as indicated by the opt[' weights'] argument. It first configures the logging and decides whether to execute the model on a CPU or GPU. It also sets the GPU if the device is one.

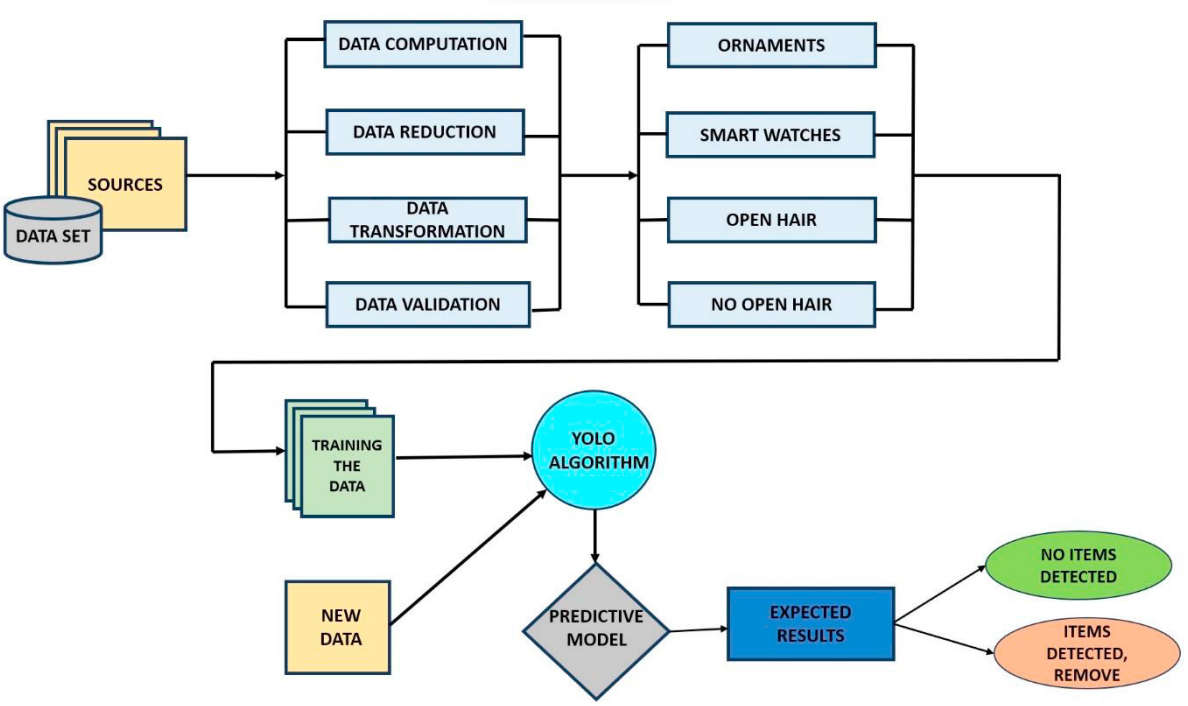


Fig. 2. Approach-for-extending-the-extraction-model.

5.4 Predictions on the Input Image

The image is then loaded into Open CV and re-sized using the letterbox function to make it compatible with the model. [17] [18] Additionally, it transposes, turns the picture to a PyTorch tensor, and changes the image from BGR to RGB. The model function, which gives bounding boxes and class probabilities for the identified items, is then used to apply the YOLOv7 model to the picture. Finally, it uses non-maximum suppression to eliminate overlapping detections and then uses the plot one box function to overlay the remaining bounding boxes on the original picture. encompassing a thorough assessment of dataset quality and appropriateness. This encompasses evaluating factors such as annotation accuracy, label consistency, bounding box quality, class balance, image quality, data splitting, scrutiny of challenging cases, review of pre-processing steps, identification and handling of duplicate images, and the application of performance metrics to ensure the robustness and reliability of the dataset in the context of YOLO object detection.

5.5 Captures Frames from a Live Video Stream

Using the camera as an input, a JavaScript code block is used to produce a live video stream, grab frames from the stream, and transform them into OpenCV pictures. Additionally, it offers the ability to overlay rectangles representing bounding boxes on video streams and transform them into base64-byte strings. A JavaScript object containing a webcam picture is passed to the js to image function, which decodes the base64-encoded image, turns it into a NumPy array, and then decodes the NumPy array into an OpenCV BGR image. The bbox to bytes method transforms a rectangle's bounding box from a NumPy array into a base64-byte string that may be overlaid over a video stream. The video stream function Fig 2. specifies a JavaScript function that takes a camera as an input and produces a live video stream. It constantly pulls frames from the video stream and sends them to a callback function using the on-animation Frame method.

5.6 Detecting the Objects

Initially put up a label for the recorded video and initialized the video stream. Then, it loads a Yolov7 model and configures the inference-related parameters, such as the input picture size and confidence level.[19] [20] The program then enters a loop that records every frame from the video stream, modifies it to the proper format, and then sends it to the model for inference. Following the post-processing of the model's output, bounding boxes are delineated around the recognized items, and a non-maximum suppression technique is subsequently employed. This code may be applied to a variety of tasks, including activity identification, object tracking, and security monitoring. Enhancing detection accuracy in cameras hinges on fundamental components: cultivating a high-quality and diverse dataset, ensuring accurate annotations, and maintaining a balanced distribution of object classes.

6. Results and Discussions

The proposed system was experimentally evaluated on the Google Colab platform and obtained an overall accuracy of 86.0%. This signifies the attainment of the highest recorded rate through numerous experiments conducted with a limited dataset, employing a batch size of 50 and 100 iterations for epochs. It was demonstrated in Figures 3 and 4. Figure 3 illustrates that the system achieves an accuracy of 0.80 for detecting a person wearing a bracelet, 0.58 for earrings, and 0.65 for open hair. Figure 4 illustrates that the system achieves an accuracy of 0.84 for detecting a person wearing earrings, and 0.86 for smart watches. At the examination centers, the camera will be activated, and if any anomalies are detected, it will display the corresponding item that has been identified. If there is any detection, the system shows a warning" Entry Denied!!!" or else the camera will run. The precision support table corresponding to the result is depicted in Table 1. With only a limited dataset, achieving such accuracy is notable. However, it is anticipated that utilizing thousands or millions of datasets could lead to even higher system accuracy.

Table 1. Precision support table of the systems

Method	Precision	Recall	F_1 Score	Support
with_openhair	0.90	0.82	0.86	11
without_openhair	0.83	0.91	0.87	11
macro average	0.87	0.86	0.86	22
weighted average	0.87	0.86	0.86	22

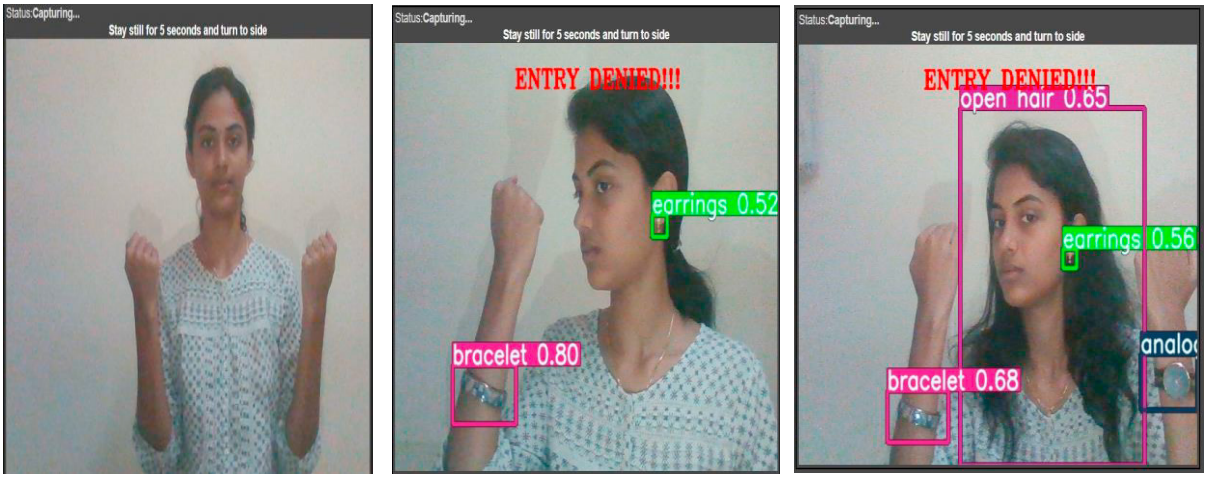


Fig. 3. Detection of open hair, earring, and bracelet through live camera

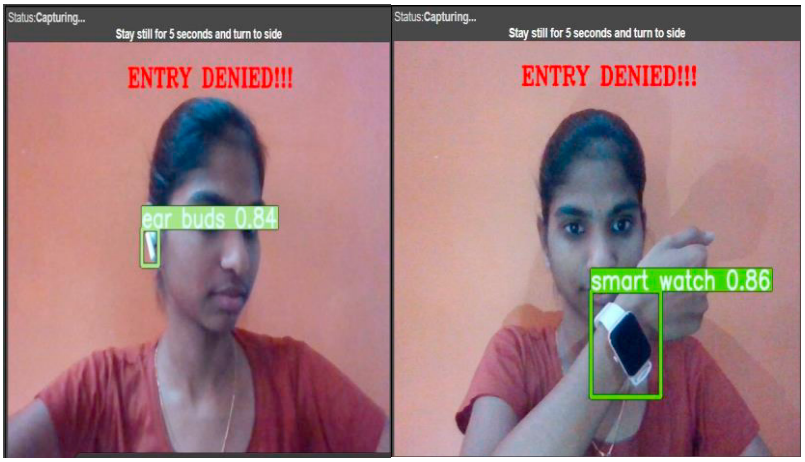


Fig. 4. Detection of ear buds and smartwatch through live camera

7. Conclusion and Future Scope

We initiated the paper by crafting our own dataset, a meticulous process that involved annotating the gathered data. Subsequently, and employed YOLOv7 to train the custom dataset, refining the model through this specialized data. Following the training, then conducted data pre-processing and loaded the customized pre-trained model. Here proceeded to make predictions on frames captured from real-time video streaming, using the trained model to detect objects. In cases where unauthorized objects were identified, a warning message—' ENTRY DENIED '—was promptly displayed.

Elevating object detection performance in a camera involves pivotal steps: validating the training dataset to mirror the diverse scenarios the camera may face, spanning different lighting conditions, object poses, and occlusions. Further improvements come through meticulous hyperparameter optimization, the application of post-processing techniques, and harnessing hardware accelerators to expedite inference speed. Enhance the system by incorporating additional features, ensuring an augmentation of capabilities while maintaining high accuracy, thereby facilitating examination procedures in strict adherence to norms and eliminating the need for manual intervention. In the scenario of offline examinations, this strategy involves developing a prototype that grants access to the examination hall only to individuals who impeccably adhere to all prescribed norms. Conversely, in the context of online examinations, the system is designed to seamlessly redirect individuals to a designated webpage for test-taking, ensuring access is granted only to those who meticulously adhere to the examination norms.

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